Course Name: Marketing Analytics

Course Code: DS 4115

Lab Manual 1: E-Commerce Cohort & Metric Analysis using Excel

Lab Overview

In this lab, you will acquire or generate a large-scale raw E-Commerce transaction dataset (100k+ rows) and perform analysis in Microsoft Excel for Windows. Using a combination of XLOOKUP and Advanced Pivot Tables, you will enrich transaction logs and calculate key business metrics, including Cohort-Level Revenue Streams, Average Order Value (AOV), and Customer Acquisition Cost (CAC).

Part 0: Step-by-Step Dataset Acquisition & Setup Guide

Why Standard Kaggle Datasets Lack "Acquisition Channel"

Public e-commerce datasets (such as Kaggle's Online Retail II) record transactional fields (InvoiceNo, CustomerID, Date, StockCode, Quantity) captured by Point-of-Sale (POS) software. Marketing attribution data (Acquisition Channel) is captured separately by web analytics platforms. In real-world enterprise environments, analysts must bridge these data silos using lookup functions.

You have two practical approaches to acquire the complete dataset for this lab:

Option A: Generate the Exact Dataset via Python (Recommended)

Run the following script locally to instantly create three CSV files matching the lab schema perfectly:

```
import pandas as pd
import random
from datetime import datetime, timedelta

random.seed(42)

# 1. Products Master Table
product_ids = [f"P{str(i).zfill(3)}" for i in range(1, 51)]
product_names = [f"Product {i}" for i in range(1, 51)]
unit_prices = [round(random.uniform(10.0, 250.0), 2) for _ in range(50)]
```

```
pd.DataFrame({
    "Product ID": product_ids,
    "Product Name": product_names,
    "Unit Price": unit_prices
}).to_csv("Products.csv", index=False)

# 2. Marketing Spend Master Table
channels = ["Google Ads", "Meta Ads", "Organic Search", "Email", "Affiliates"]
pd.DataFrame({
    "Acquisition Channel": channels,
    "Total Spend": [25000, 30000, 5000, 10000, 15000],
    "New Customers Acquired": [1200, 1500, 800, 900, 1100]
}).to_csv("Marketing_Spend.csv", index=False)

# 3. 100,000+ Rows Transaction Dataset
num_rows = 105000
start_date = datetime(2025, 1, 1)

pd.DataFrame({
    "Transaction ID": [f"TXN{str(i).zfill(7)}" for i in range(1, num_rows + 1)],
    "Customer ID": [f"CUST{random.randint(1000, 9999)}" for _ in range(num_rows)],
    "Date": [(start_date + timedelta(days=random.randint(0, 364))).strftime("%Y-%m-%d") for _ in
range(num_rows)],
    "Product ID": [random.choice(product_ids) for _ in range(num_rows)],
    "Units Sold": [random.randint(1, 5) for _ in range(num_rows)],
    "Acquisition Channel": [random.choice(channels) for _ in range(num_rows)]
}).to_csv("Transactions.csv", index=False)

print("Dataset generated successfully!")
```

Option B: Download Real Data from Kaggle & Map Columns in Excel

- Download the **Online Retail II** dataset from Kaggle or the UCI Machine Learning Repository.
- Open the file in Excel and rename the headers to standard names:
    - InvoiceNo to Transaction ID
    - CustomerID to Customer ID
    - InvoiceDate to Date
    - StockCode to Product ID
    - Quantity to Units Sold
- Add the missing **Acquisition Channel** column by entering this formula in cell F2 and copying it down:

```
=INDEX({"Google Ads","Meta Ads","Organic Search","Email","Affiliates"}, RANDBETWEEN(1,5))
```

- Copy the entire Acquisition Channel column and **Paste as Values** (Ctrl + Alt + V -> Values) to lock the data.

Part 1: Dataset Ingestion & Column Enrichment (XLOOKUP)

Step 1: Prepare the Workbook

- Save your dataset as an Excel Workbook (.xlsx).
- Ensure you have three worksheets: Transactions, Products, and Marketing_Spend.

Step 2: Fetch Unit Price using XLOOKUP

- On the **Transactions** sheet, add header **Unit Price** in cell G1.
- Enter the formula in G2:

```
=XLOOKUP(D2, Products!$A$2:$A$51, Products!$C$2:$C$51, 0)
```

- Double-click the fill handle to apply the formula across all rows.

Step 3: Compute Gross Revenue & Cohort Month

- Add header **Gross Revenue** in cell H1. In H2, enter:

```
=E2 * G2
```

- Add header **Cohort Month** in cell I1. In I2, enter:

```
=TEXT(C2, "YYYY-MM")
```

- Copy both formulas down the entire dataset.

Part 2: Channel-Level CAC Analysis (XLOOKUP)

Step 1: Calculate CAC on Marketing Sheet

- Open the **Marketing_Spend** sheet.
- Add header **CAC** in column D.
- In cell D2, enter:

```
=B2 / C2
```

- Format column D as Currency ($).

Step 2: Link CAC to Main Transactions Sheet

- Return to the **Transactions** sheet and add header **Channel CAC** in cell J1.
- Enter the formula in J2:

```
=XLOOKUP(F2, Marketing_Spend!$A$2:$A$6, Marketing_Spend!$D$2:$D$6, 0)
```

Part 3: Advanced Pivot Table Cohort Analysis

Step 1: Create the Pivot Table

- Select any cell inside the Transactions dataset and press **Ctrl + A**.
- Go to **Insert** > **PivotTable** > **New Worksheet** > **OK**.

Step 2: Configure Pivot Table Fields

- Drag **Cohort Month** to **Rows**.
- Drag **Acquisition Channel** to **Columns**.
- Drag **Gross Revenue** to **Values** (Sum of Gross Revenue).
- Drag **Transaction ID** to **Values** (Count of Transaction ID).
- Drag **Gross Revenue** to **Values** a second time.

Step 3: Format Values for Insights

- **AOV:** Right-click the second Gross Revenue field -> **Summarize Values By** -> **Average**. Rename to AOV ($).
- **Revenue Share:** Right-click the third Gross Revenue field -> **Value Field Settings** -> **Show Values As** -> **% of Column Total**. Rename to % Channel Revenue.

Step 4: Add Interactive Slicers

- Click inside the Pivot Table.
- Go to **PivotTable Analyze** > **Insert Slicer**.
- Select **Acquisition Channel** and **Cohort Month**, then click **OK**.

Deliverables Checklist for Students

Ingested/generated 100k+ rows of transactional data saved as .xlsx.
Enriched dataset with Unit Price, Gross Revenue, Cohort Month, and Channel CAC using XLOOKUP.
Calculated channel-level Customer Acquisition Cost (CAC).
Built multi-dimensional Pivot Table analyzing total revenue, AOV, and channel percentage split.
Added interactive Slicers for cohort analysis.